THE LOCUS WHERE A LOCAL SECTION OF AN ETALE SHEAF VANISHES IS THE LARGEST ZARISKI OPEN SUBSET AND IS DETECTED BY GEOMETRIC POINTS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the three numbered parts, each written out inside the counted span.

Editorial note. The label lemma-zero-over-image occurs in no other staged Stacks chapter, so no cross-chapter restatement had to be weighed for this card.

Editorial note. The single labelled reference occurring inside the counted spans points to another chapter of the Project; it is carried in assembly.external_references as a non-load-bearing navigation pointer recorded at the invoking line.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Restriction of an etale sheaf to the Zariski site of a scheme. Same chapter, the fact that a sheaf for the etale topology restricts to a sheaf for the Zariski topology on U. Supplies the Zariski sheaf whose vanishing locus W is produced in part (1).
- Sheaf condition and etale coverings V -> phi(V). Same chapter, the sheaf condition for F on U_etale applied to the etale covering {V -> phi(V)}. Supplies the step that a section vanishing on V vanishes on the image phi(V), which proves part (2).
- Stalk of an etale sheaf at a geometric point and etale neighbourhoods. Same chapter, the description of the stalk F_sbar as classes of sections over etale neighbourhoods, with vanishing meaning vanishing on some etale neighbourhood. Supplies the equivalence in part (3) between vanishing at a geometric point and containment of that point in W.

Source reference. Original lemma 01AU: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3969–3969. Cross-chapter navigation pointer invoked at proof lines 3968-3969, where the author uses that the zero locus of a section of a sheaf on a scheme is Zariski open. The target chapter is not staged in this card's source tree, so the pointer records the invoking line; it only resolves the reference for the reader, and no bibliography citation is involved.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{F}$ be an abelian sheaf on $S_{étale}$. Let $\sigma \in \mathcal{F}(U)$ be a local section. There exists an open subset $W \subset U$ such that*

(1) *$W \subset U$ is the largest Zariski open subset of U such that $\sigma|_W = 0$,*

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(2) *for every* $\varphi : V \to U$ *in* $S_{étale}$ *we have*
$$\sigma|_V = 0 \Leftrightarrow \varphi(V) \subset W,$$
(3) *for every geometric point* $\overline{u}$ *of* U *we have*
$$(U, \overline{u}, \sigma) = 0 \text{ in } \mathcal{F}_{\overline{s}} \Leftrightarrow \overline{u} \in W$$
where $\overline{s} = (U \to S) \circ \overline{u}$.

Proof. Since $\mathcal{F}$ is a sheaf in the étale topology the restriction of $\mathcal{F}$ to U_{Zar} is a sheaf on U in the Zariski topology. Hence there exists a Zariski open W having property (1), see Modules, Lemma 01AU. Let $\varphi : V \to U$ be an arrow of $S_{étale}$. Note that $\varphi(V) \subset U$ is an open subset and that $\{V \to \varphi(V)\}$ is an étale covering. Hence if $\sigma|_V = 0$, then by the sheaf condition for $\mathcal{F}$ we see that $\sigma|_{\varphi(V)} = 0$. This proves (2). To prove (3) we have to show that if $(U, \overline{u}, \sigma)$ defines the zero element of $\mathcal{F}_{\overline{s}}$, then $\overline{u} \in W$. This is true because the assumption means there exists a morphism of étale neighbourhoods $(V, \overline{v}) \to (U, \overline{u})$ such that $\sigma|_V = 0$. Hence by (2) we see that $V \to U$ maps into W, and hence $\overline{u} \in W$. □